Lecture on Conformal Prediction (4/4)

Margaux Zaffran

October 29-30, 2026

LIDAM – ISBA seminar series

FONDATION MATHÉMATIQUE JACQUES HADAMARD

Laboratoire de Mathématiques d'Orsay

LIDAM | Louvain Institute of Data Analysis and Modeling in economics and statistics

ISBA

Last lecture's topic

1. On exchangeability (theory)
2. Split conformal prediction (methods) (theory) (practical session)
3. Towards conditional coverage? (practical session) (theory) (case studies)
4. Beyond exchangeability (methods) (case studies)
5. Computational and statistical trade-offs (methods) (theory)

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5. Computational and statistical trade-offs (methods) (theory)
6. Handling missing data (methods)

Handling missing data

- Supervised learning setting with missing covariates
- Goals and challenges for predictive uncertainty quantification
- Is MCV a too lofty goal?!
- Achieving MCV under $M \perp\!\!\!\perp X$ and $Y \perp\!\!\!\perp M \,|X$
- Experimental results

A collaboration



Yaniv Romano

Technion - Israel Institute of Technology



Julie Josse

PreMeDICaL

INRIA



Aymeric Dieuleveut

École Polytechnique

- *Predictive Uncertainty Quantification with Missing Covariates*, 2024
- *Conformal Prediction with Missing Values*, ICML 2023

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- More than 30 000 trauma patients
- 4 000 new patients per year
- 250 continuous and categorical variables
  $\hookrightarrow$ Many useful statistical tasks

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These covariates are not always observed.

Missing values are ubiquitous and challenging

Data: $\left(X^{(k)}, \quad Y^{(k)}\right)_{k=1}^{n}$

Y	X_1	X_2	X_3
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19	6	8	NA
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$$d = 6 \rightarrow \approx 94\% \text{ of rows kept}$$

$$d = 300 \rightarrow \approx 5\% \text{ of rows kept}$$

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One of the **ironies of Big Data** *is that missing data play an ever more significant role.*[1]

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$\hookrightarrow M$ can depend on X or Y (depending on the missing mechanism[1]).

[1] Three mechanisms connecting X and M from **?**, *Inference and missing data*, Biometrika

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$\Rightarrow$ Statistical and computational challenges.

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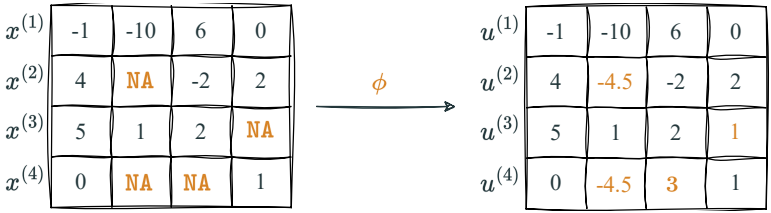
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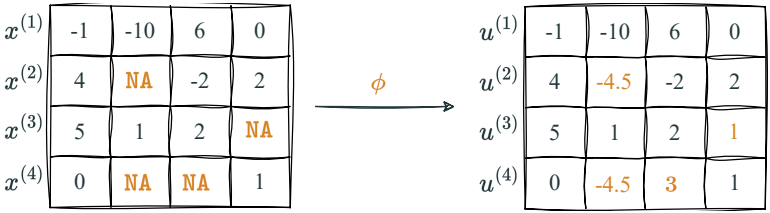
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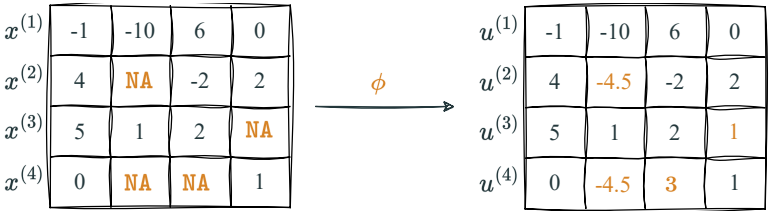


2. Train your algorithm (Random Forest, Neural Nets, etc.) on the imputed data: $\left\{ \underbrace{\phi\left(X^{(k)}_{\text{obs}(M^{(k)})}, M^{(k)}\right)}_{U^{(k)}=\text{imputed } X^{(k)}}, Y^{(k)} \right\}_{k=1}^{n}$.

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$\hookrightarrow$ we consider an impute-then-predict pipeline in this work.

Supervised learning with missing values: impute-then-predict, cont'd

✓ Le Morvan et al. (2021)[2] show that for **any deterministic imputation** and **universal learner** this procedure is **Bayes-consistent**.

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- ✓ Le Morvan et al. (2021)[2] show that for any deterministic imputation and universal learner this procedure is Bayes-consistent.
- ✗ Ayme et al. (2022)[3] show that even for very **simple distributions** (linear model, Gaussian noise), this rate of convergence may suffer from **curse of dimensionality**.

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Goals of predictive uncertainty quantification with missing values

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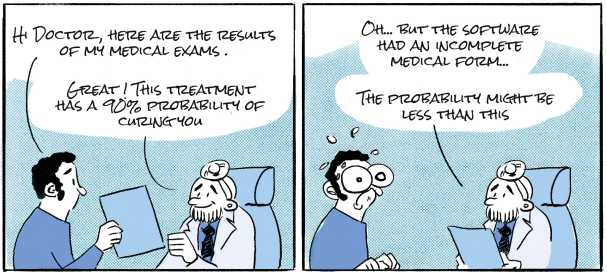
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Definition (2. Mask-Conditional-Validity (MCV))**.**

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Illustrations @theoremlinger

CP is marginally valid (MV) after imputation

Lemma (Exchangeability after imputation (Z., Dieuleveut, Josse and Romano, 2023)).

Assume $\left(X^{(k)}, M^{(k)}, Y^{(k)}\right)_{k=1}^{n}$ are i.i.d. (or exchangeable).
Then, for any missing mechanism, for almost all imputation function ϕ:
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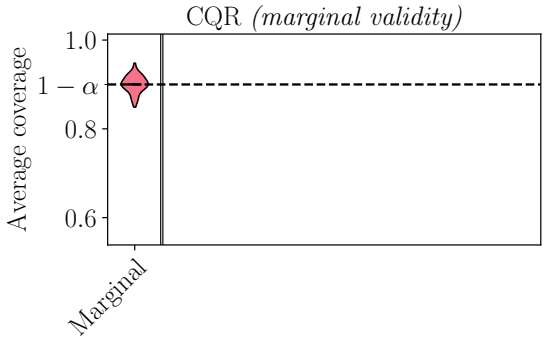
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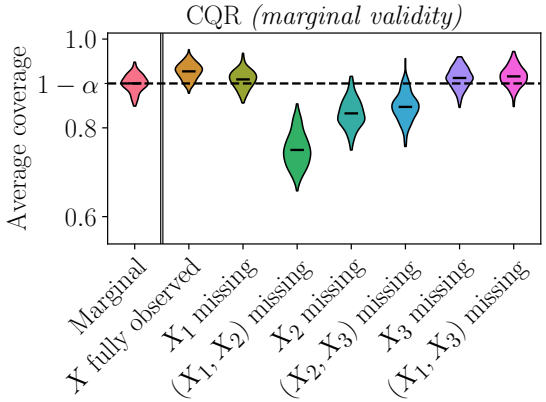
$Y = \beta^T X + \varepsilon$, $\beta = (1, 2, -1)^T$, X and ε Gaussian.



✓ Marginal (i.e. average) coverage (MV) is indeed recovered!

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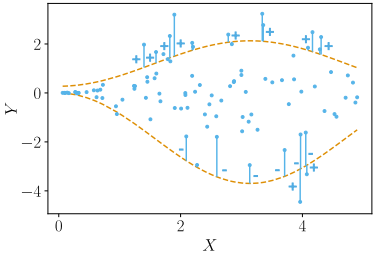
✗ Mask-conditional-validity (MCV) is not attained

↪ Missing values induce heteroskedasticity

(supported by theory under (non-)parametric assumptions)

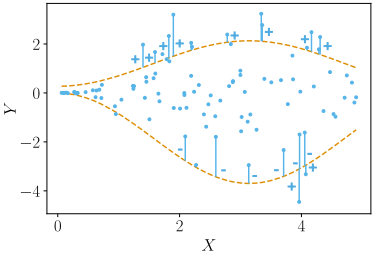
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Warning: 2^d possible masks

$\Rightarrow$ Splitting the calibration set by mask is infeasible (lack of data)!

Initial calibration set

$x^{(1)}$	-1	-10	6	1
$x^{(2)}$	4	NA	-2	2
$x^{(3)}$	5	1	1	NA
$x^{(4)}$	0	NA	NA	1

Test point

3	6	0	1

Calibration set used

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Conceptually: a structured distribution shift situation

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3. Reasonable model of the link between pattern and the uncertainty.

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Theorem (General MCV hardness result)**.**

If $\widehat{C}_\alpha$ is distribution-free MCV then for any distribution $\mathcal{D}$, for any mask m such that $\mathcal{D}_M(m) := \mathbb{P}_{\mathcal{D}}(M = m) > 0$, it holds:

- *Regression:*
$$\mathbb{P}_{\mathcal{D}^{\otimes(n+1)}}\left(\text{mes}\left(\widehat{C}_\alpha\left(X_{n+1}, m\right)\right) = \infty\right) \underset{\text{if } \mathcal{D}_M(m) \ll 1/n}{\simeq} 1 - \alpha$$
- *Classification:*
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Irreducible term: consider $\widehat{C}_\alpha$ outputting $\mathcal{Y}$ with probability $1-\alpha$ and $\emptyset$ otherwise.

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$$\mathbb{P}_{\mathcal{D}^{\otimes(n+1)}}\left(\text{mes}\left(\widehat{C}_\alpha\left(X_{n+1}, m\right)\right) = \infty\right) \underset{\text{if } \mathcal{D}_M(m) \ll 1/n}{\simeq} 1 - \alpha$$
- *Classification:*
for any $y \in \mathcal{Y}$, $\mathbb{P}_{\mathcal{D}^{\otimes(n+1)}}\left(y \in \widehat{C}_\alpha\left(X_{n+1}, m\right)\right) \underset{\text{if } \mathcal{D}_M(m) \ll 1/n}{\simeq} 1 - \alpha$

Irreducible term: consider $\widehat{C}_\alpha$ outputting $\mathcal{Y}$ with probability $1 - \alpha$ and $\emptyset$ otherwise.

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Theorem (General MCV hardness result)**.**

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$\Rightarrow$ need to restrict both the link between M and X, as well as between M and Y

Handling missing data

- Supervised learning setting with missing covariates
- Goals and challenges for predictive uncertainty quantification
- Is MCV a too lofty goal?!
- Achieving MCV under $M \perp\!\!\!\perp X$ and $Y \perp\!\!\!\perp M \,|X$
- Experimental results

CP-MDA-Nested* (Missing Data Augmentation)

Idea: for each test point, modify the calibration points to mimic the test mask

$\mapsto$ Solution 1: **Missing data augmentation**.

Test point

$x^{(n+1)}$

1	NA	NA	2

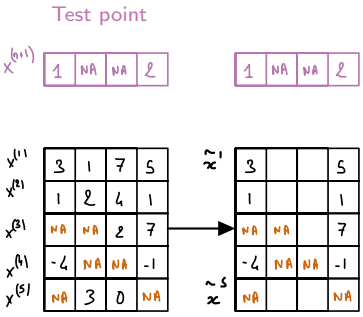
$x^{(1)}$	3	1	7	5
$x^{(2)}$	1	2	4	1
$x^{(3)}$	NA	NA	2	7
$x^{(4)}$	-4	NA	NA	-1
$x^{(5)}$	NA	3	0	NA

Calibration set

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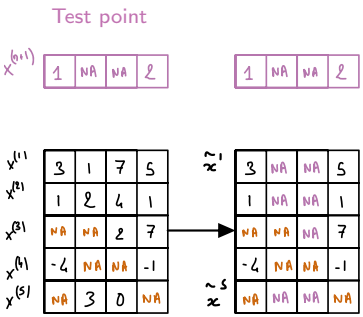
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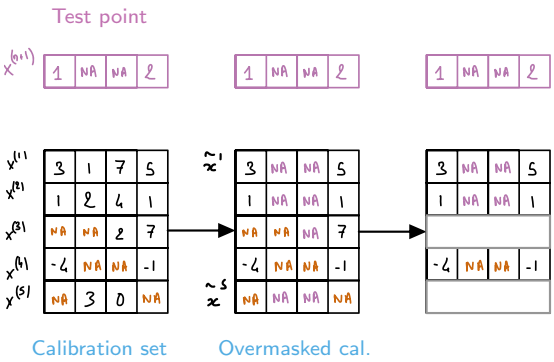
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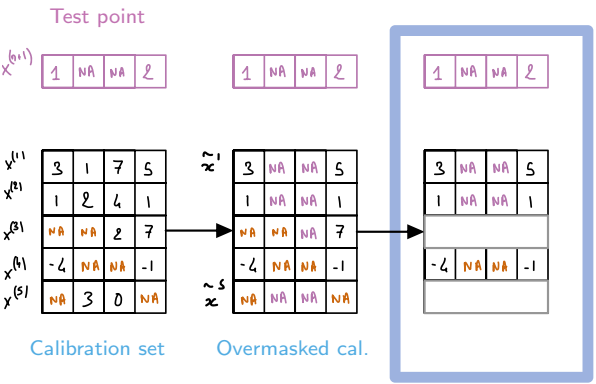
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Exchangeable!

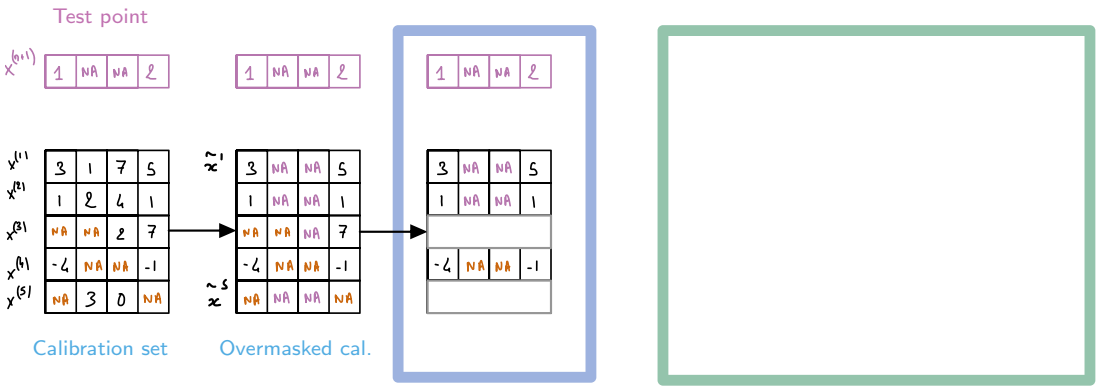
Marginal distribution

$=$ dist of $(X^{(n+1)}, M^{(n+1)}, Y^{n+1})|M^{n+1}$

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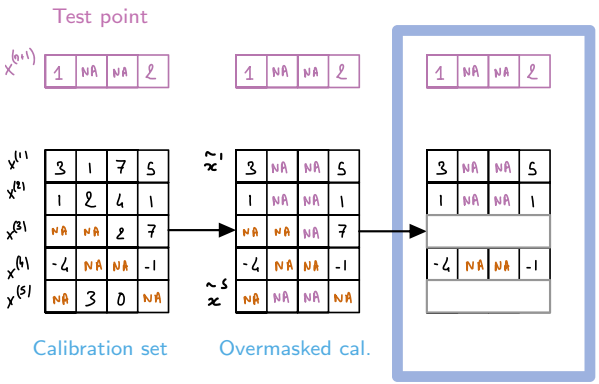
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$$\mathbb{P}\left\{Y^{(n+1)} \notin \mathcal{C}_\alpha\left(X^{(n+1)}, m\right) | M^{(n+1)}\right\} \overset{a.s.}{\leq} \alpha.$$

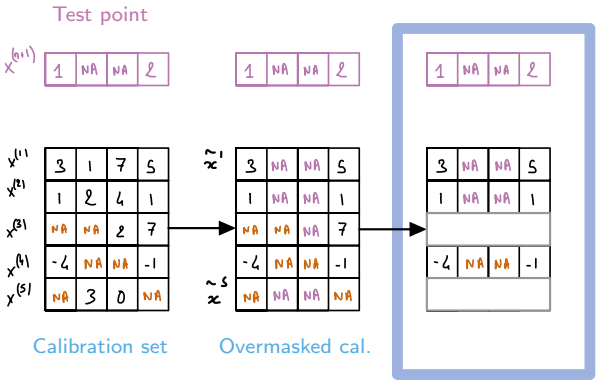
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$=$ dist of $(X^{(n+1)}, M^{(n+1)}, Y^{n+1})|M^{n+1}$

$\rightarrow$ Any CP algorithm ensures MCV under $Y \perp\!\!\!\perp M \,|X$

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$\rightarrow$ Any CP algorithm ensures MCV under $Y \perp\!\!\!\perp M \,|X$

"Nickname": MDA-Exact

Theorem (CP-MDA-Exact achieves MCV)**.**

If:

i) the data is exchangeable,

ii) $M \perp\!\!\!\perp X$,

iii) $(Y \perp\!\!\!\perp M)|X$,

then for almost all imputation function CP-MDA-Exact is such that for any $m \in \{0,1\}^d$:

$$\mathbb{P}\left(Y \in \widehat{C}_\alpha(X, m) \,|\, M = m\right) \geq 1 - \alpha,$$

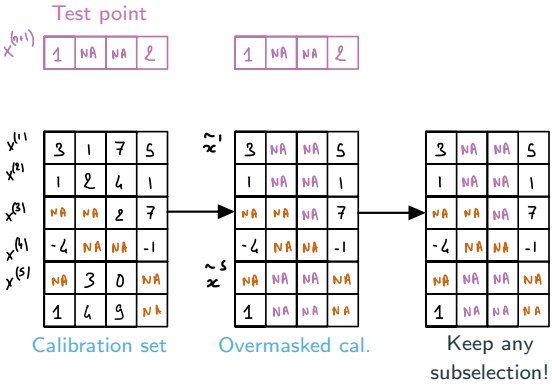
and if additionally the scores are almost surely distinct:

$$\mathbb{P}\left(Y \in \widehat{C}_\alpha(X, m) \,|\, M = m\right) \leq 1 - \alpha + \frac{1}{\#\mathrm{Cal}^m + 1}.$$

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Idea: for each test point, modify the calibration points to mimic the test mask

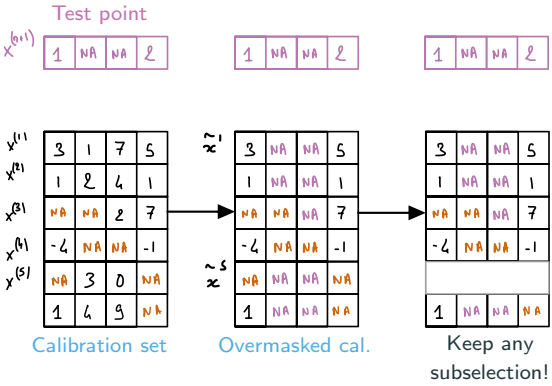
$\mapsto$ Solution 2: **Missing data augmentation + keeping more points**.



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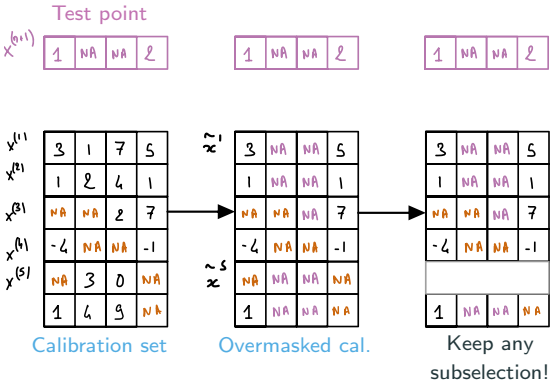
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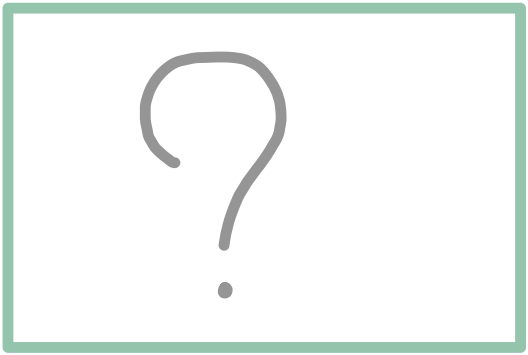


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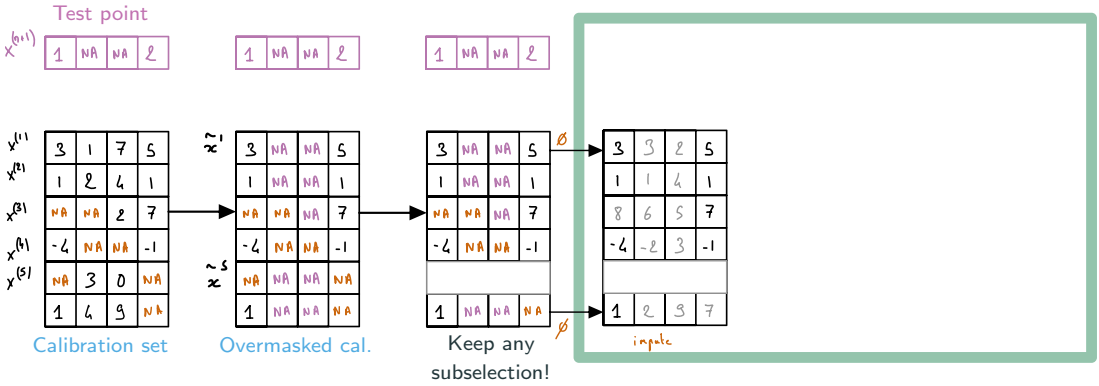




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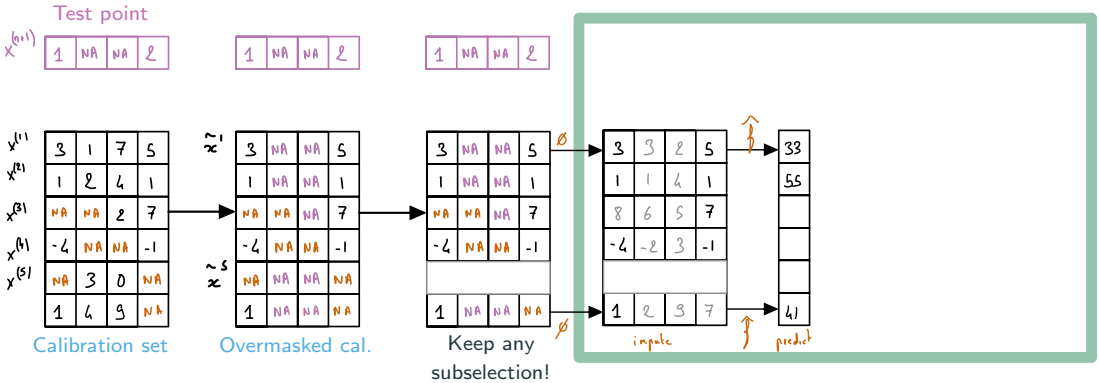
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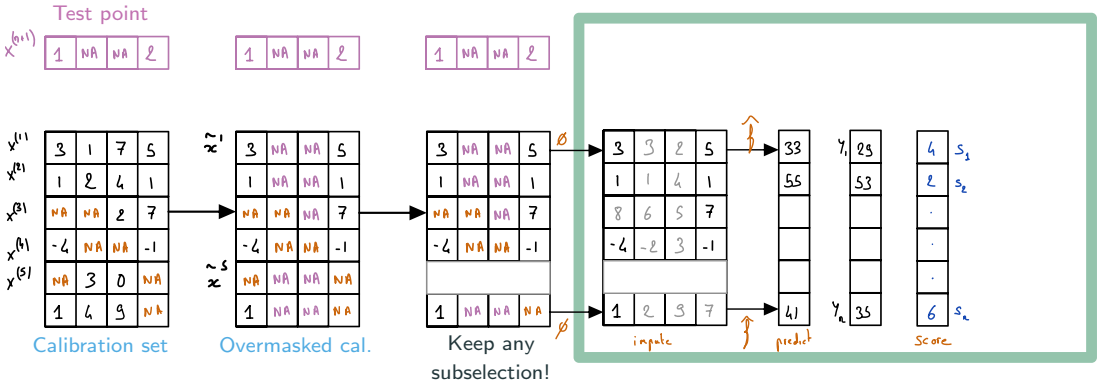
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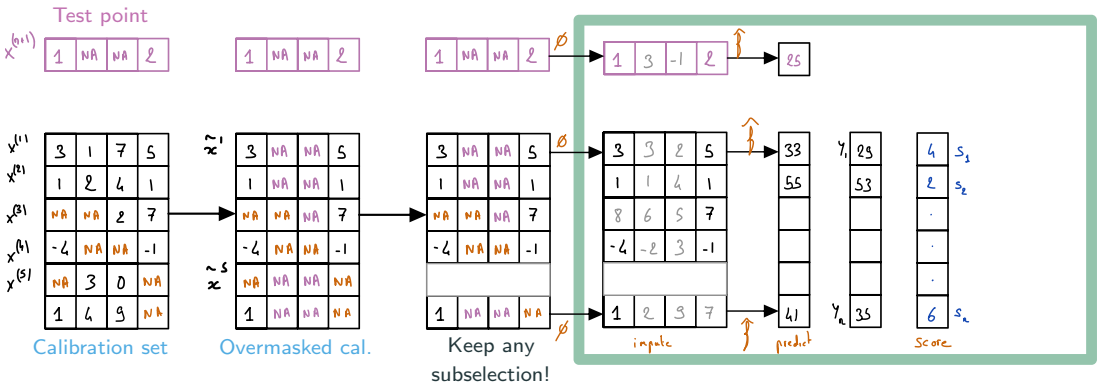
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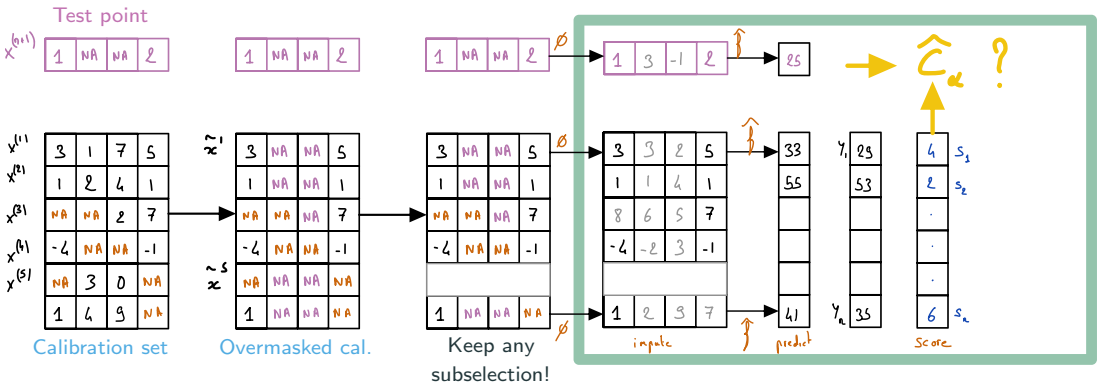
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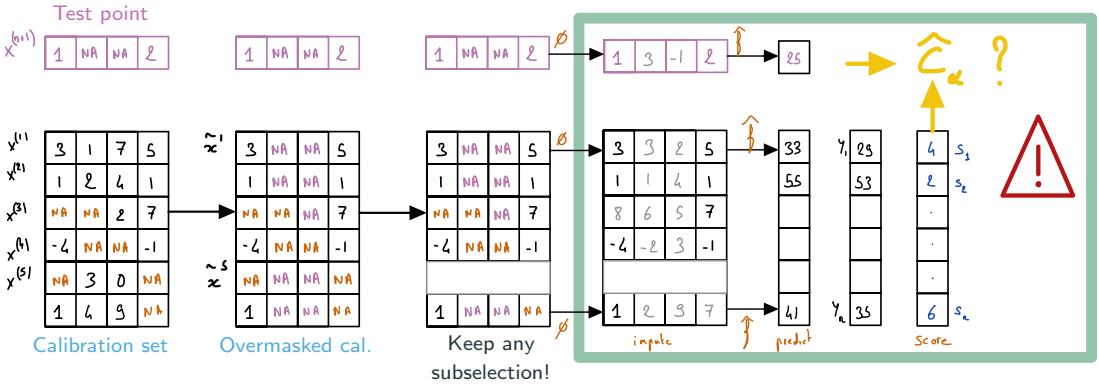
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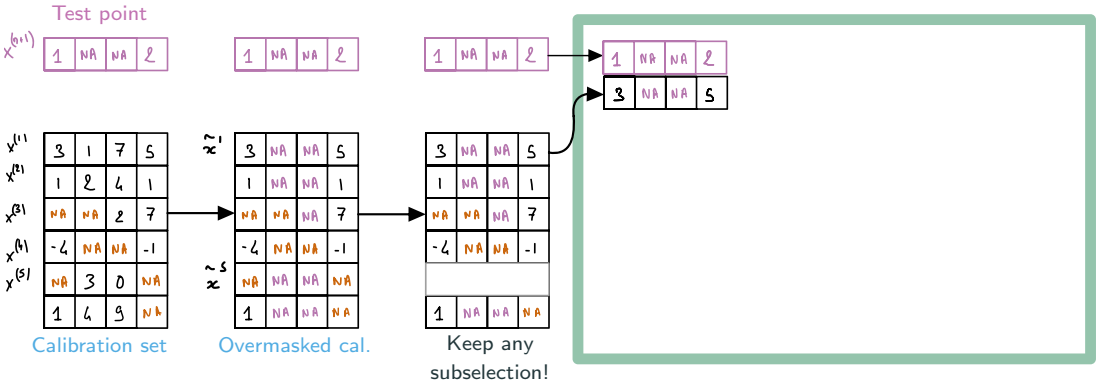


Problem: scores cannot simply be aggregated and used, no guarantee on that!

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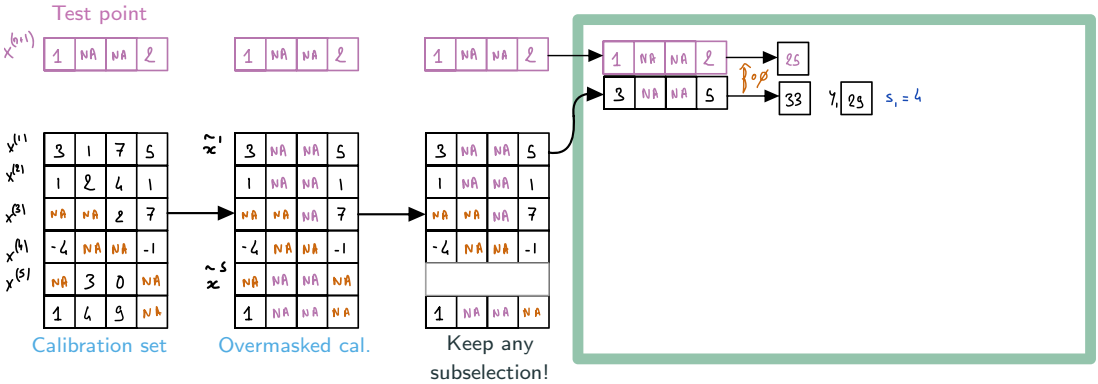
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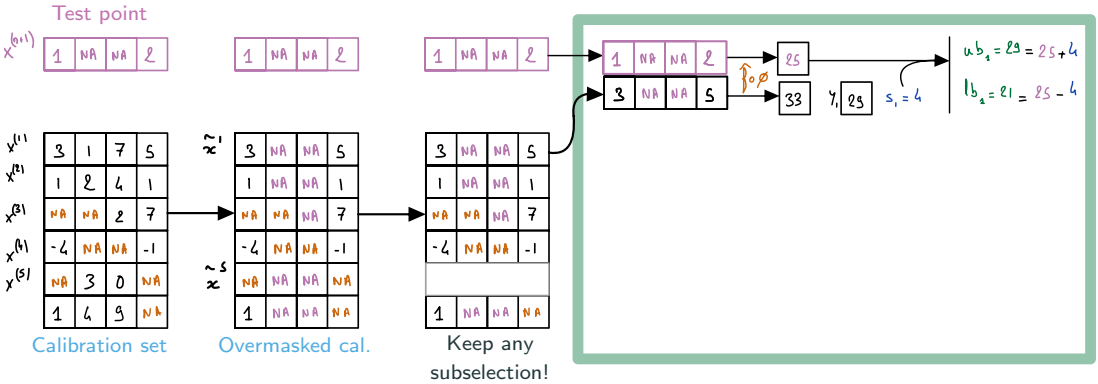
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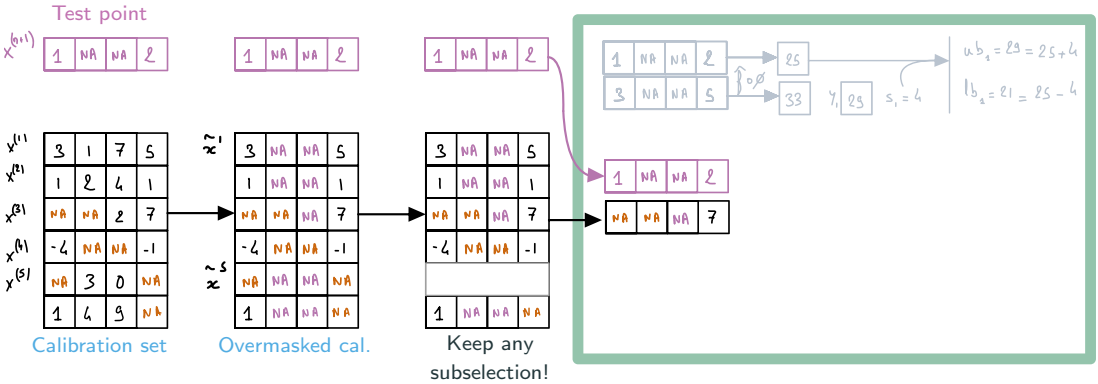
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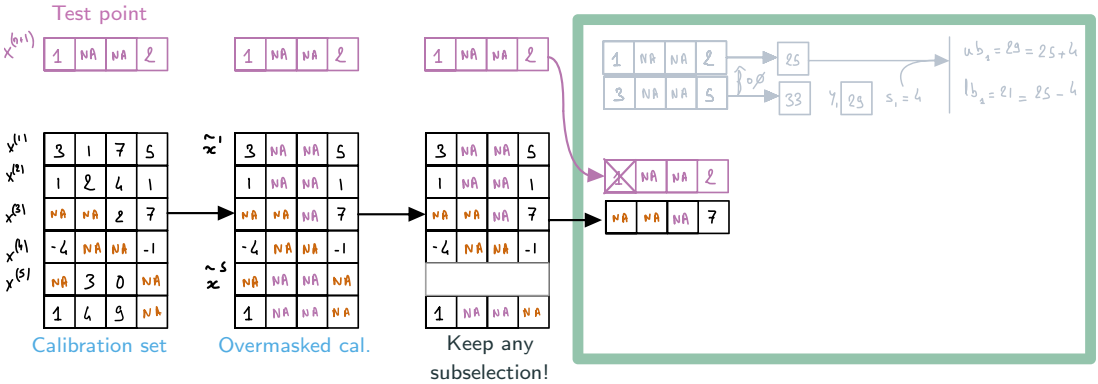
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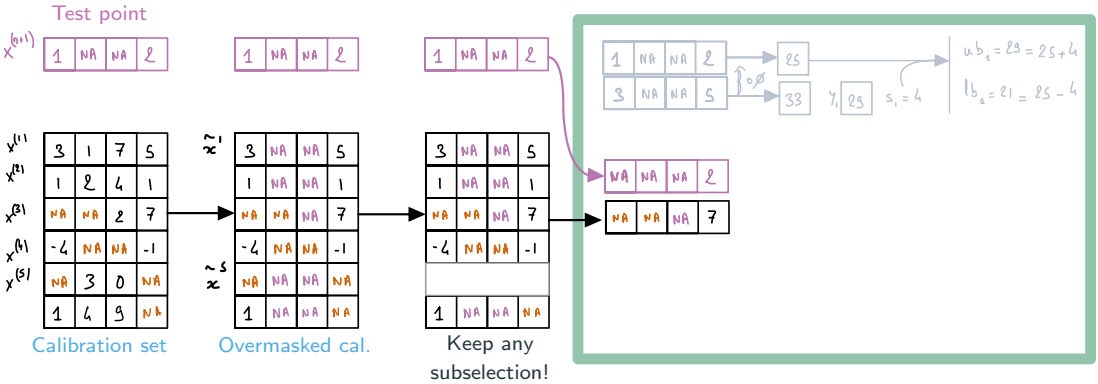
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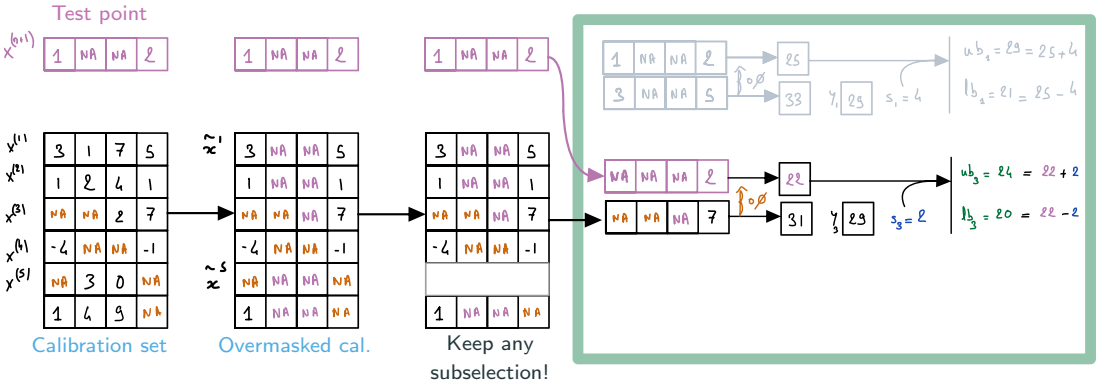
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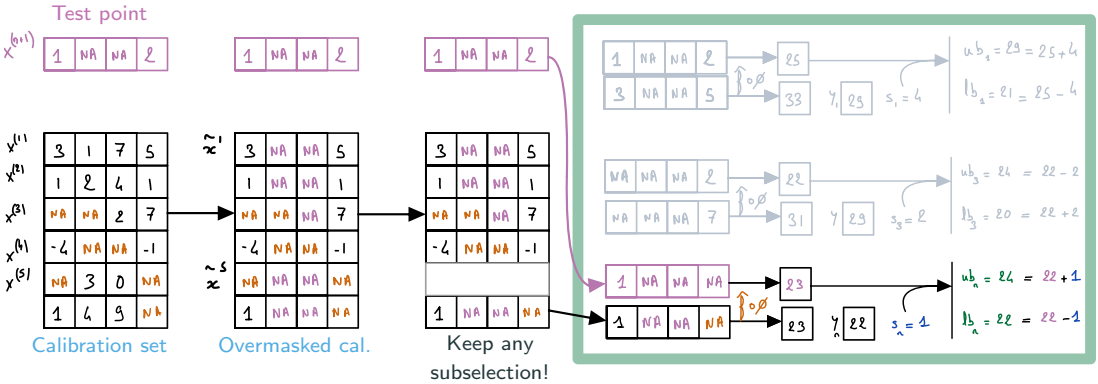
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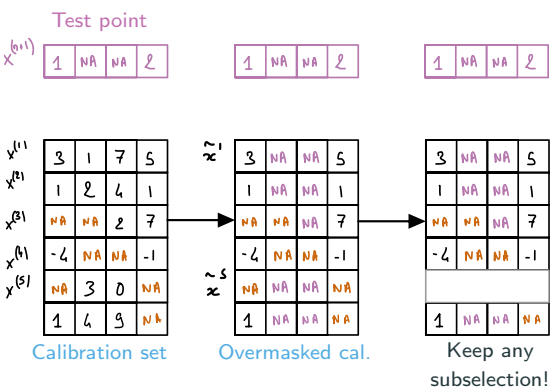
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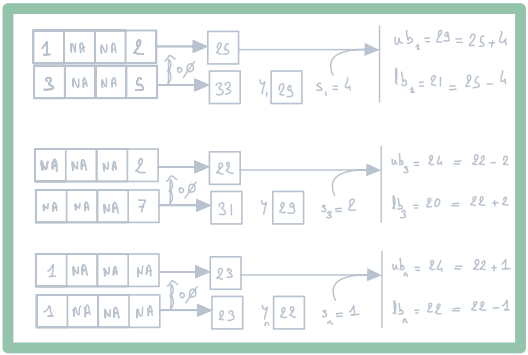


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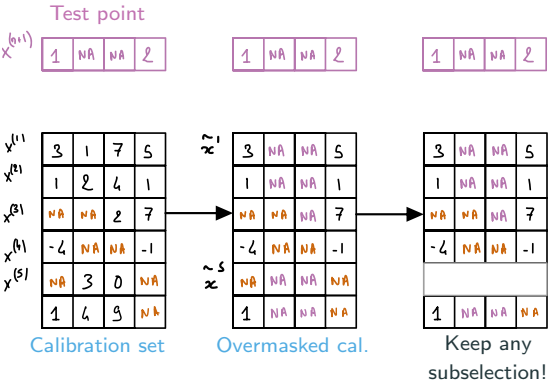
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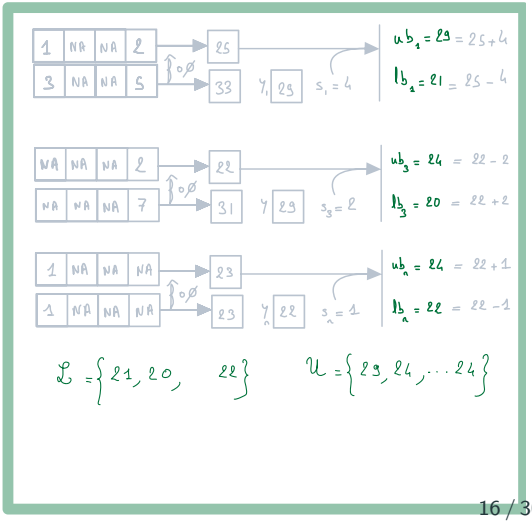




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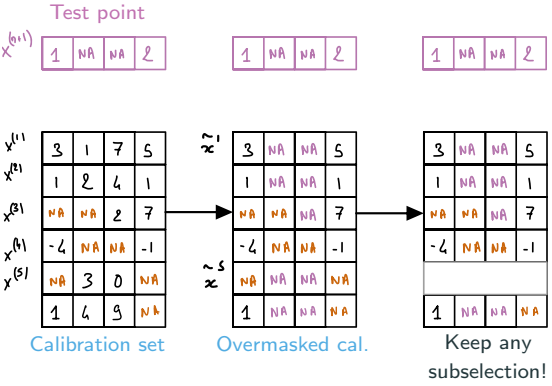
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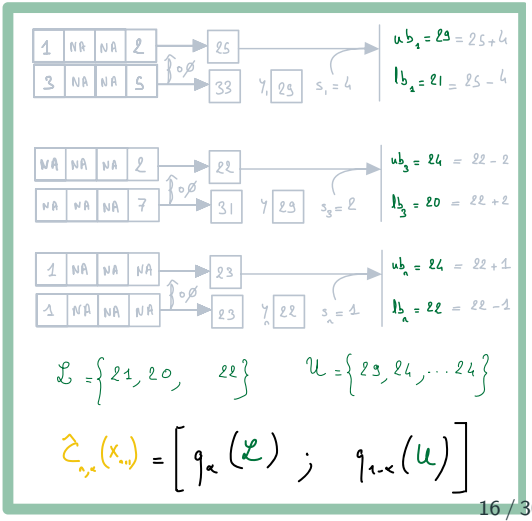




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CP-MDA-Nested* is Marginally Valid (MV)

Theorem (CP-MDA-Nested* marginal validity)**.**

If the data is exchangeable, then for almost all imputation function CP-MDA-Nested* is such that:

$$\mathbb{P}\left(Y \in \widehat{C}_\alpha\left(X, M\right)\right) \geq 1 - 2\alpha.$$

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- ✓ Does not require $(Y \perp\!\!\!\perp M) \,|\, X$
- ✗ Marginal guarantee

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Proof element: based on Jackknife+ ideas (Barber et al., 2021).

Leaving-out the k-th data point to predict on the l-th data point

$\leftrightarrow$

Apply the mask of the k-th data point to the l-th data point on which you predict

Theorem (Mask-conditional-validity of CP-MDA-Nested*)**.**

Under the assumptions that:

- $\left(X^{(k)}, M^{(k)}, Y^{(k)}\right)_{k=1}^{n+1}$ are i.i.d. with $M \perp\!\!\!\perp (X, Y)$,
- the subsampling scheme is independent of $\left(X^{(k)}, Y^{(k)}\right)_{k=1}^{n+1}$,

then, for almost all imputation function, CP-MDA-Nested* reaches (MCV) at the level $1-2\alpha$, that is:

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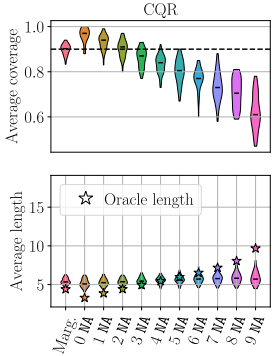
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Proof elements:

1. Crop the data sets to hide the missing entries of the test point
2. Applying the mask of the calibration point corresponds to a predictor that draws a predictions randomly

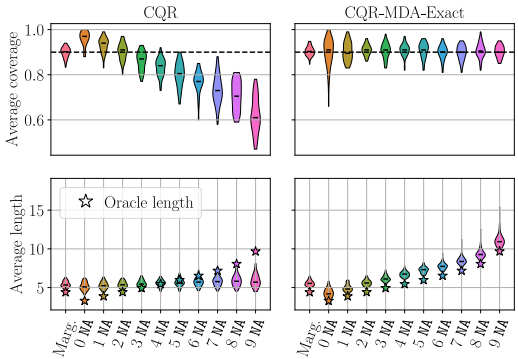
$\Rightarrow$ Use the same proof arguments than (Barber et al., 2021) on random predictors

Experiments on $M \perp\!\!\!\perp X$ and $Y \perp\!\!\!\perp M \,|X$ Gaussian linear data in dimension 10



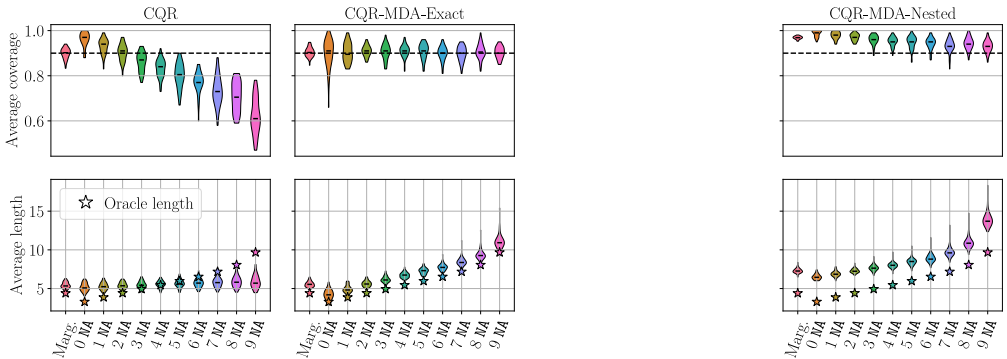
20% of missing values

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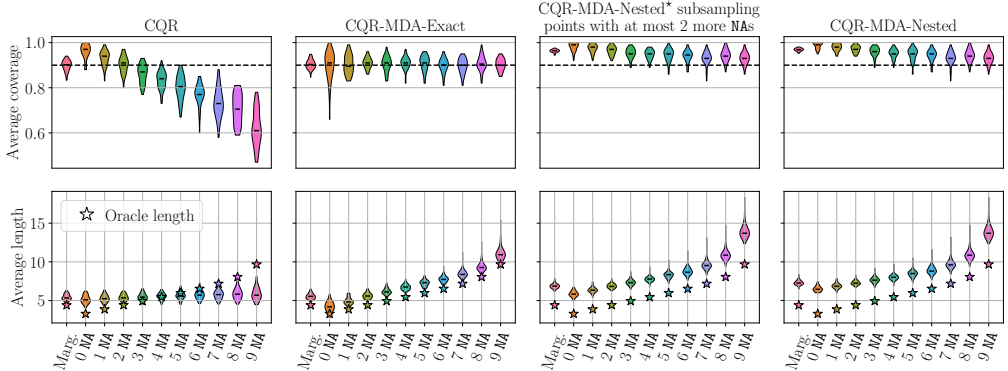
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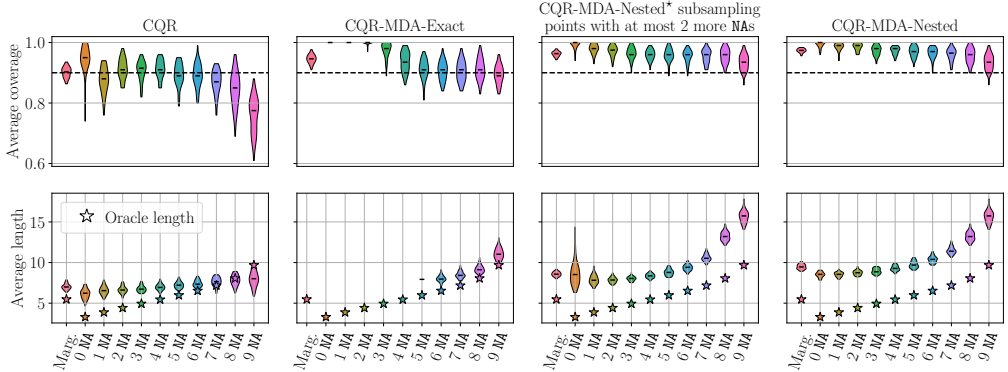
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Experiments on $M \perp\!\!\!\perp X$ and $Y \perp\!\!\!\perp M \,|X$ Gaussian linear data in dimension 10



40% of missing values

Goal: predict $Y^{(n+1)}$ with **confidence** $1-\alpha$, i.e. build the smallest $\mathcal{C}_\alpha$ such that:

Definition (1. Marginal Validity (MV)).

$$\mathbb{P}\left\{Y^{(n+1)} \in \mathcal{C}_\alpha\left(X^{(n+1)}, M^{(n+1)}\right)\right\} \geq 1-\alpha. \qquad \text{(MV)}$$

Definition (2. Mask-Conditional-Validity (MCV)).

$$\mathbb{P}\left\{Y^{(n+1)} \in \mathcal{C}_\alpha\left(X^{(n+1)}, M^{(n+1)}\right) | M^{(n+1)}\right\} \overset{a.s.}{\geq} 1-\alpha. \qquad \text{(MCV)}$$

	Exisiting approaches	
(MV)	✓ (Z., Dieuleveut, Josse, and Romano, 2023)	
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Handling missing data

- Supervised learning setting with missing covariates
- Goals and challenges for predictive uncertainty quantification
- Is MCV a too lofty goal?!
- Achieving MCV under $M \perp\!\!\!\perp X$ and $Y \perp\!\!\!\perp M \,|X$
- Experimental results

Some settings

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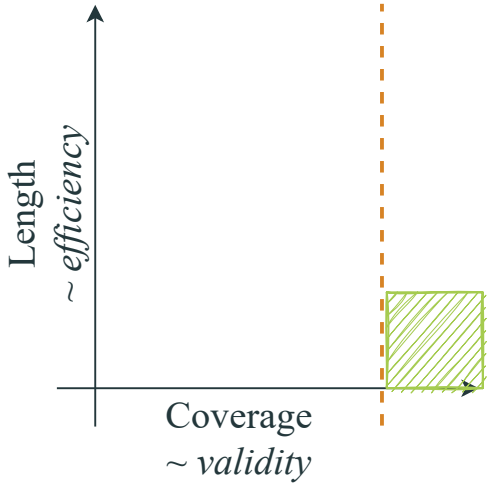
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- (Semi)-synthetic experiments:

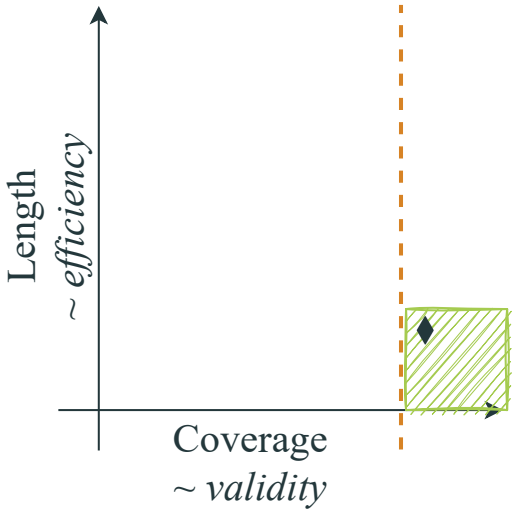
Some settings

- Imputation by iterative ridge ($\sim$ conditional expectation)
- **Concatenate the mask in the features**
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- (Semi)-synthetic experiments:
  - Uniform MCAR missing values, with probability 20%
  - 100 repetitions

Before more experiments, visualisation

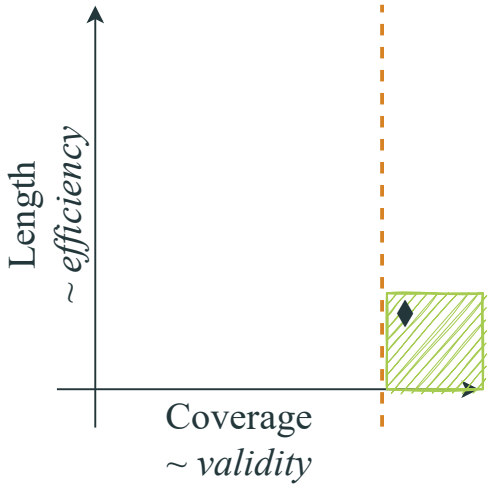


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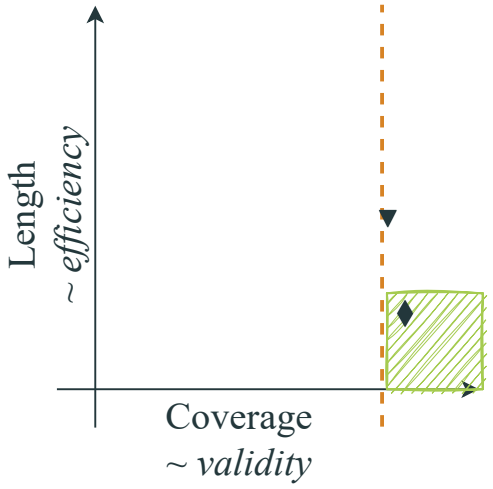
⧫ : marginal coverage, i.e.
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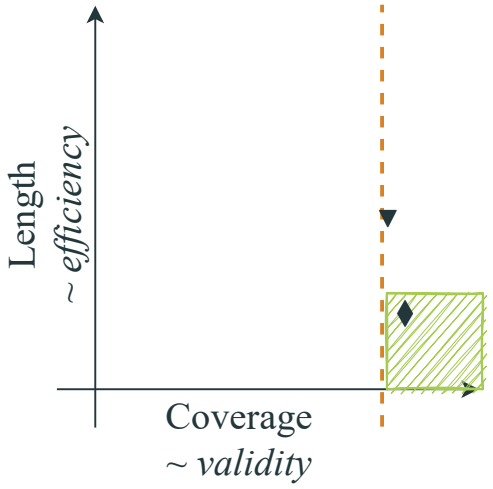
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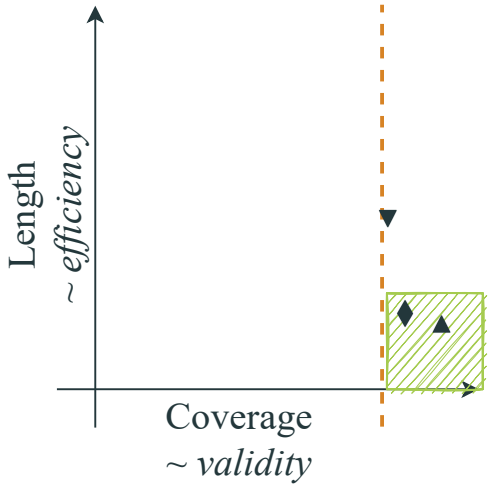
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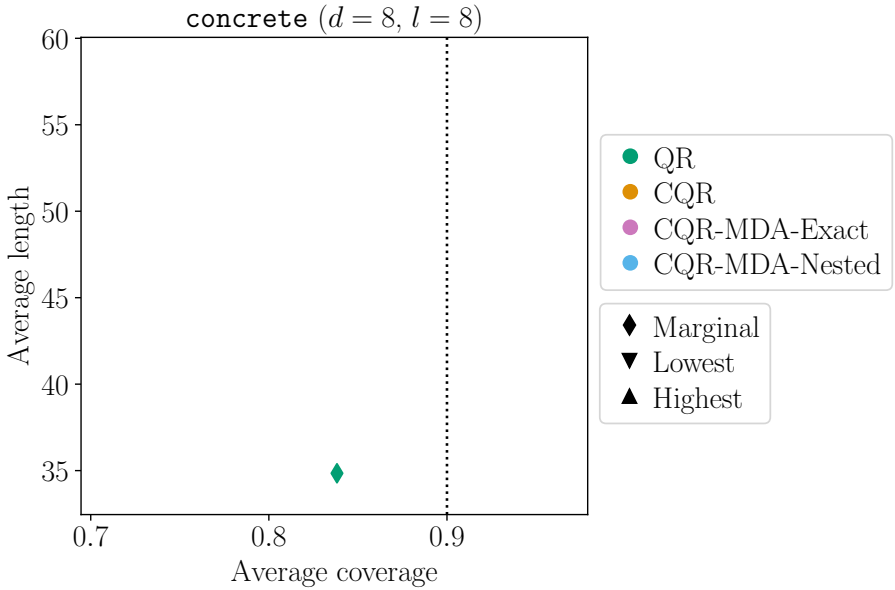
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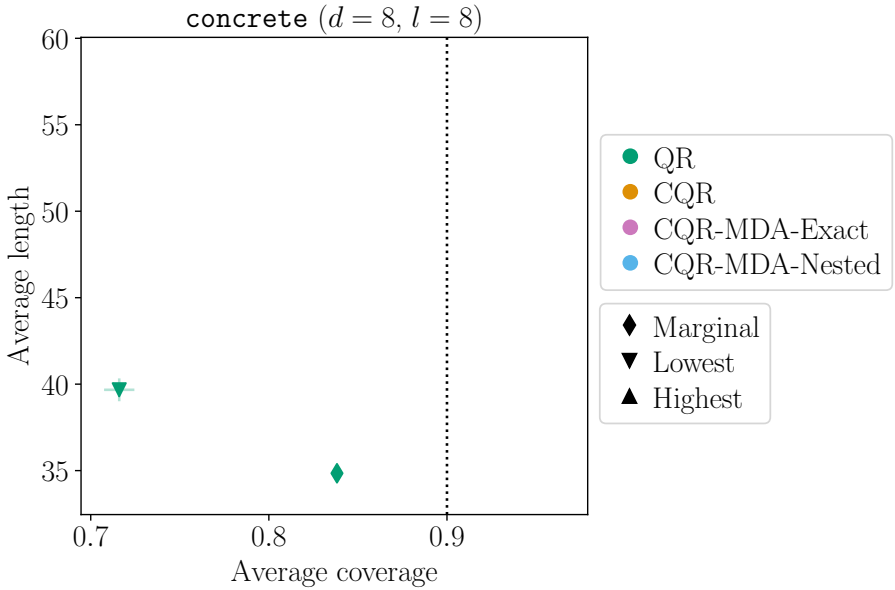


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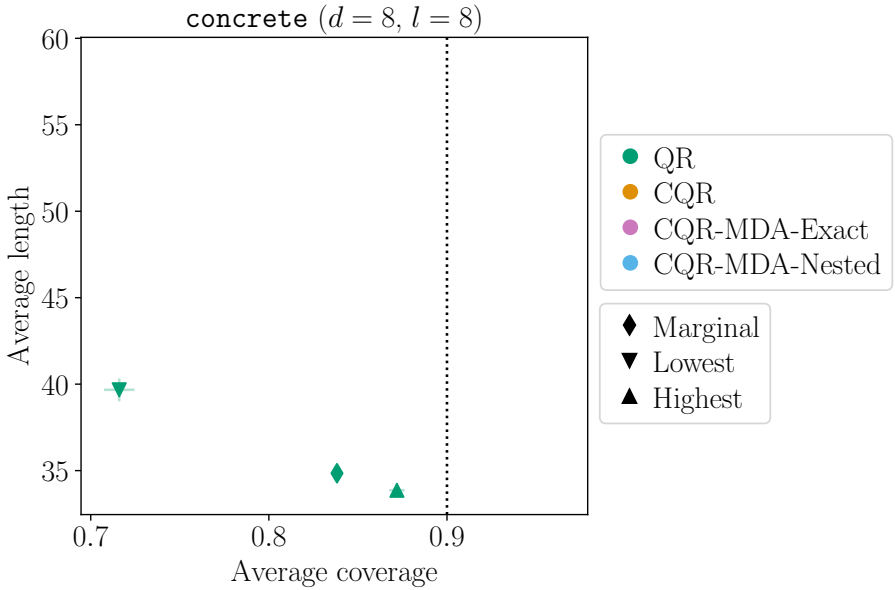
Semi-synthetic experiments



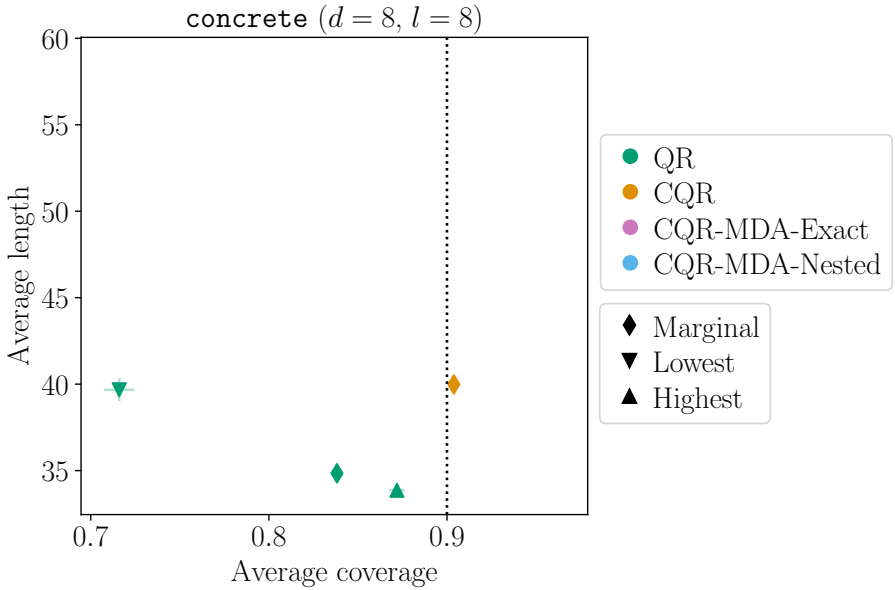
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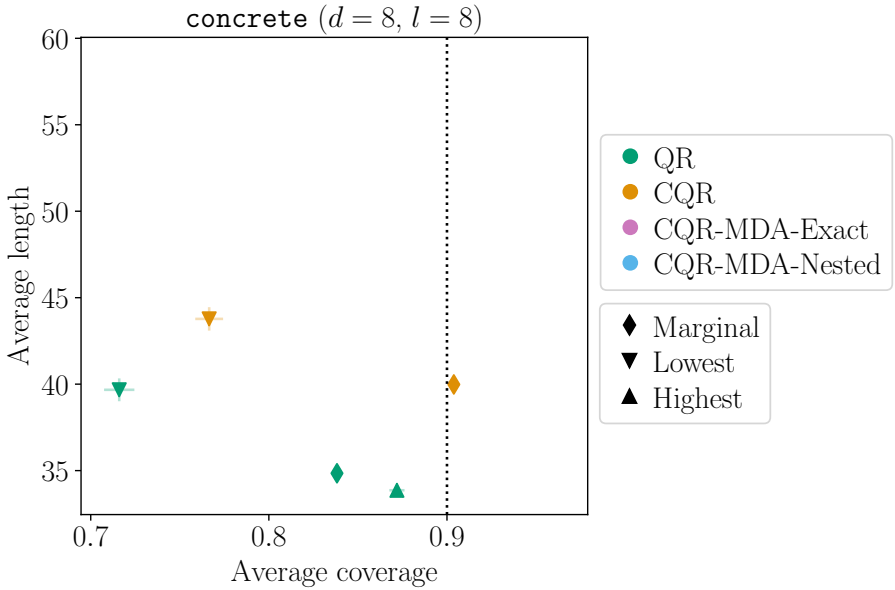
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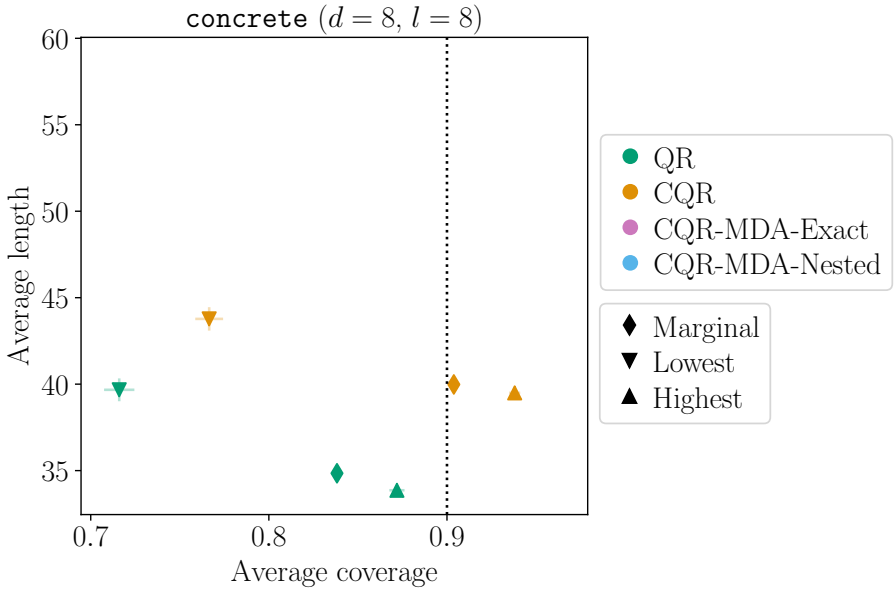
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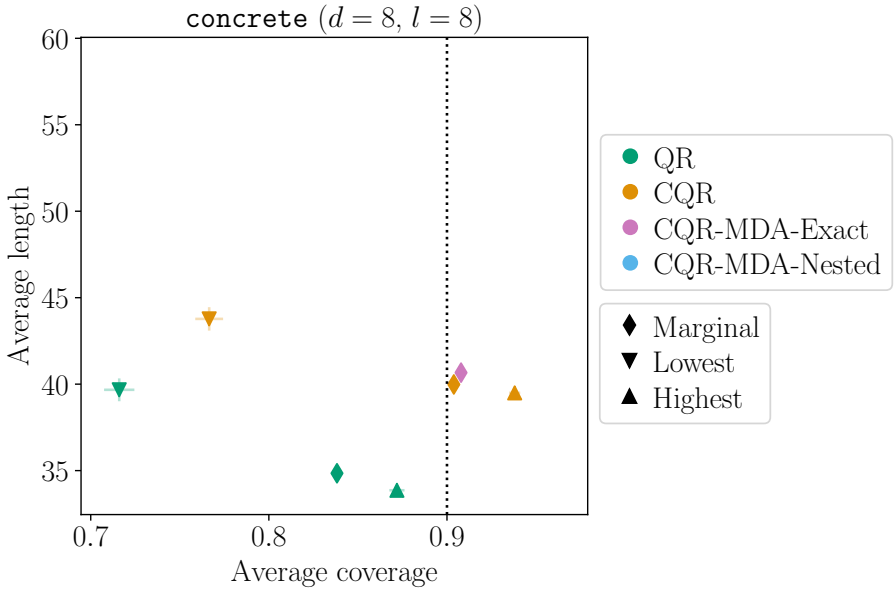
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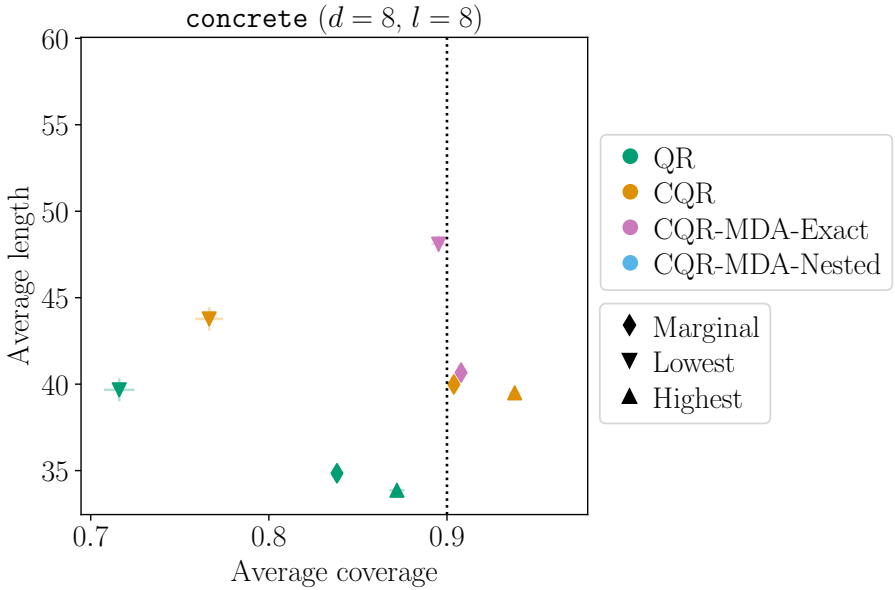
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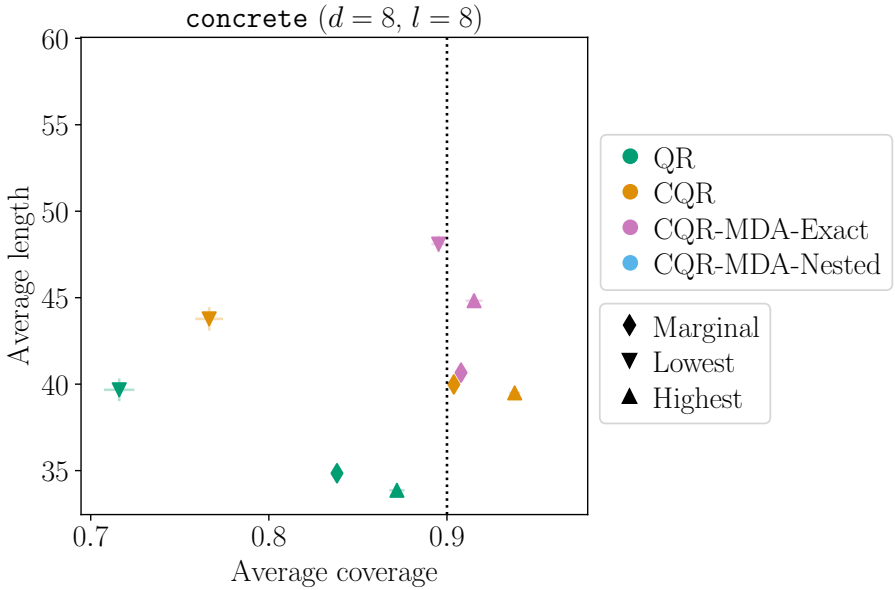
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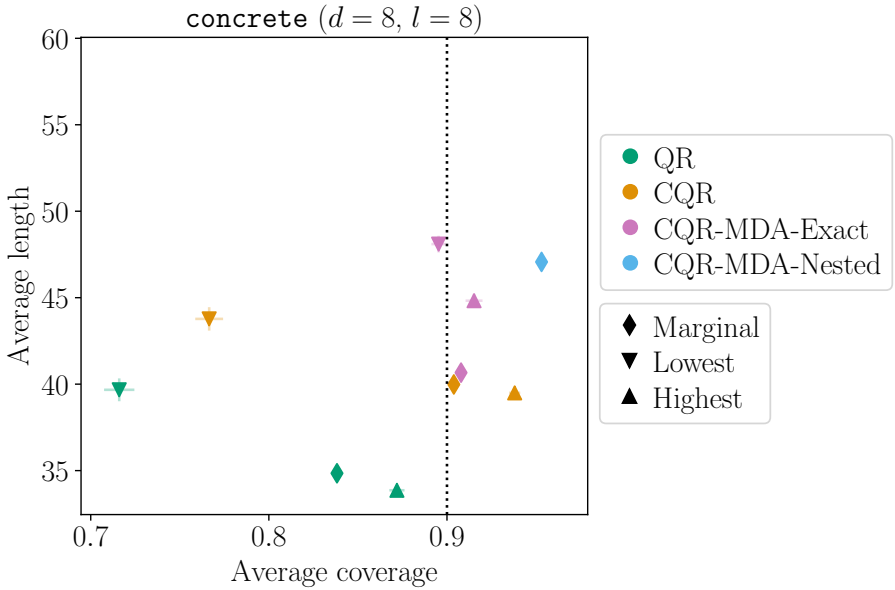
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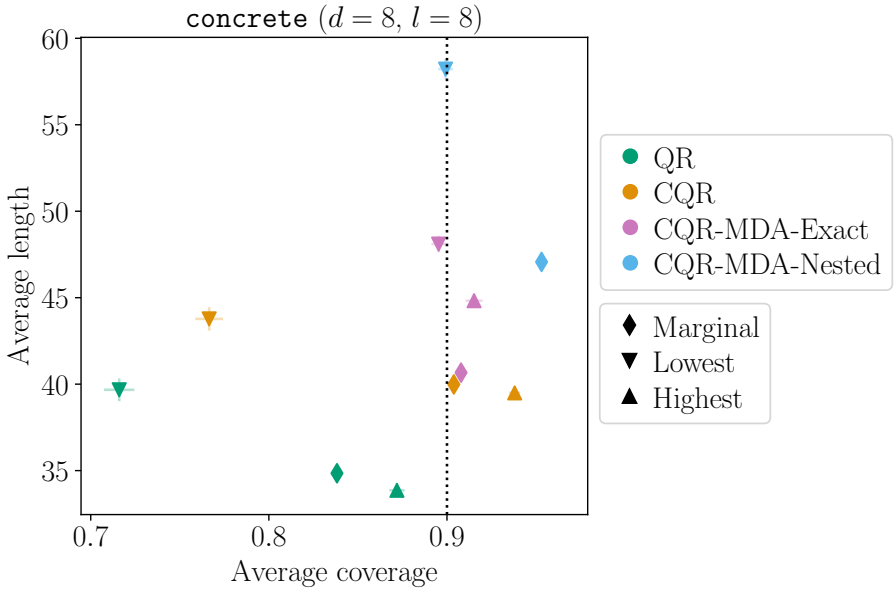
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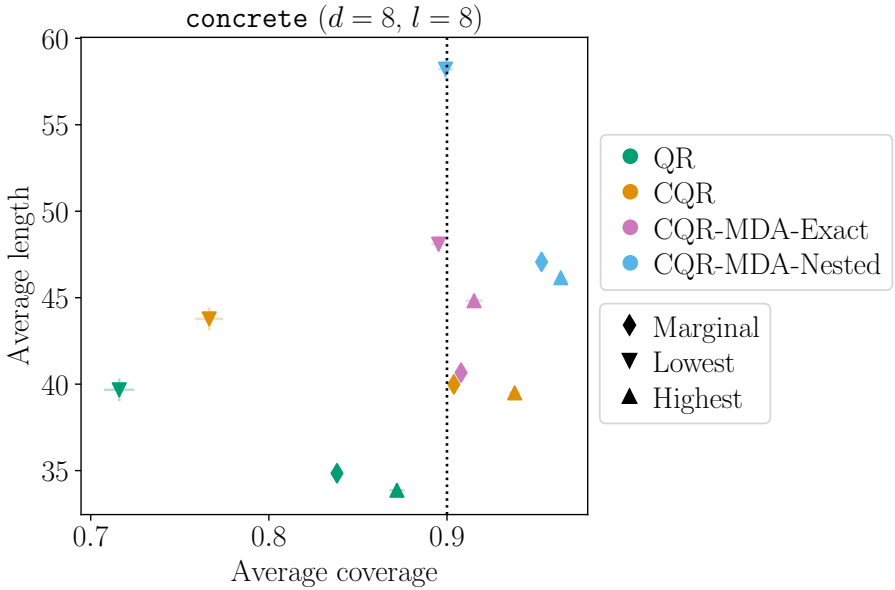
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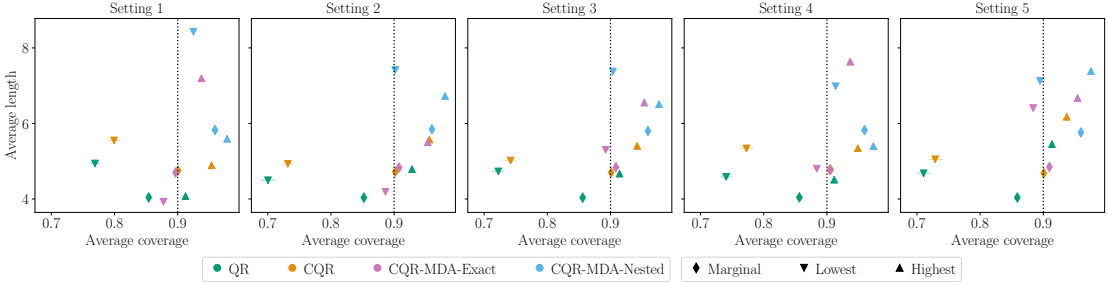
Simulation settings: beyond MCAR

- 6 variables (denote this set X_{missing}) out of 10 can be missing (the 4 others form the set X_{observed})

  $\rightarrow$ $X_{\text{missing}} = \{X_1, X_2, X_3, X_5, X_8, X_9\}$;
- Proportion of missing entries fixed to be 20%.

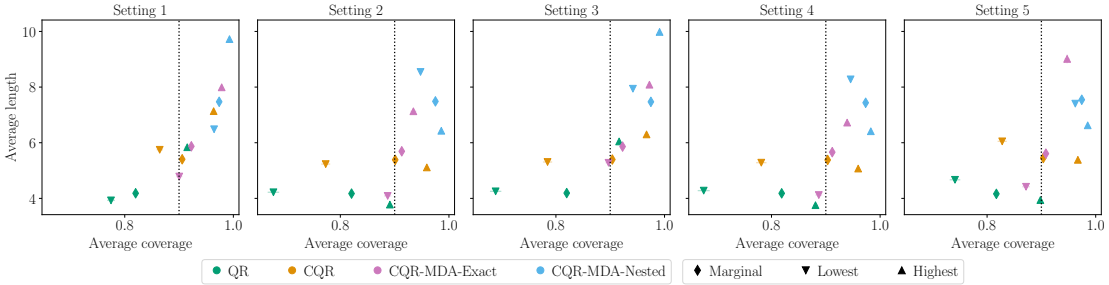
MAR missingness

- Probability of the variables in X_{missing} to be missing given by a logistic model of arguments X_{observed}.
- This setting is declined 5 times, with different weights for the logistic model.



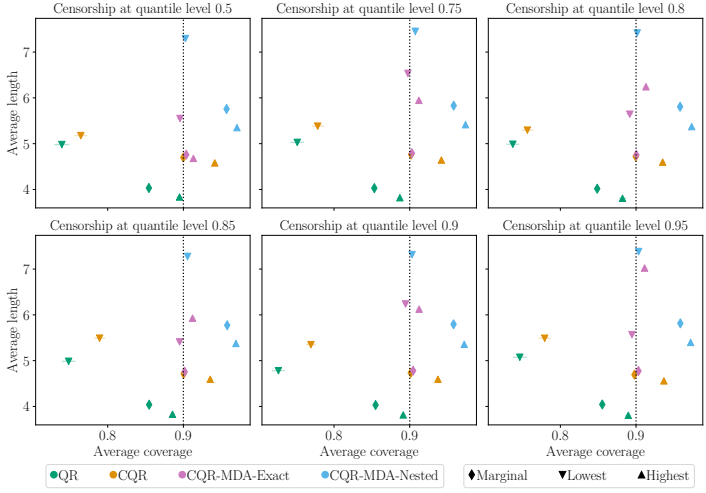
MNAR self masked missingness

- Probability of each variable in X_{missing} to be missing given by a logistic model of argument the same variable of X_{missing}.
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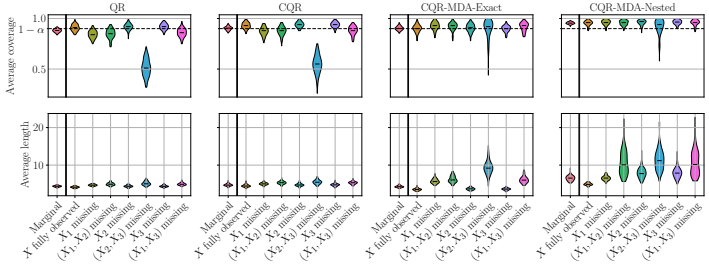
MNAR quantile censorship missingness

- Missing values are introduced at random in each q-quantile of the variables in X_{missing}.
- 6 different settings: q varies between 0.5, 0.75, 0.8, 0.85, 0.9 and 0.95.

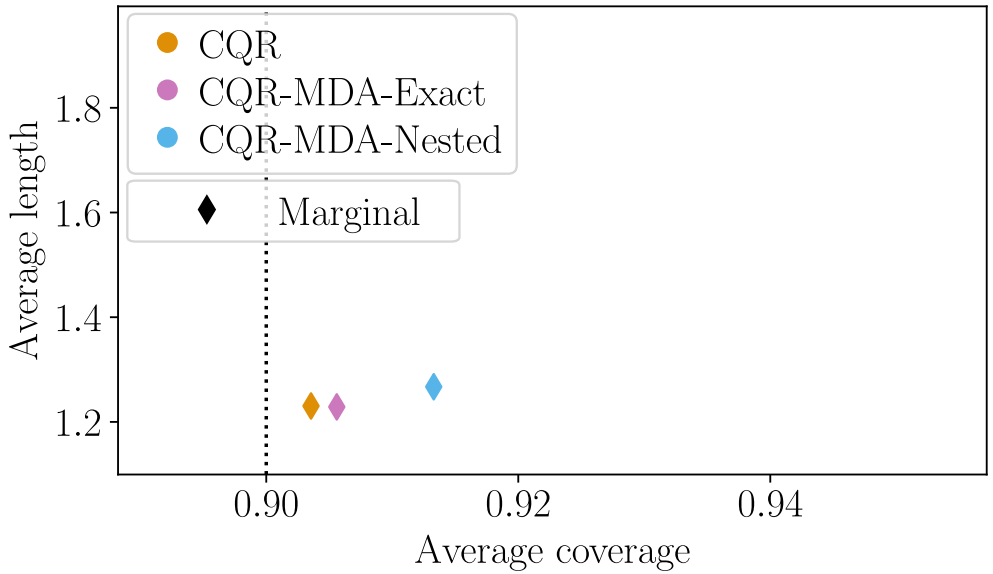


Experiments under $Y \not\perp\!\!\!\perp M \,|X$

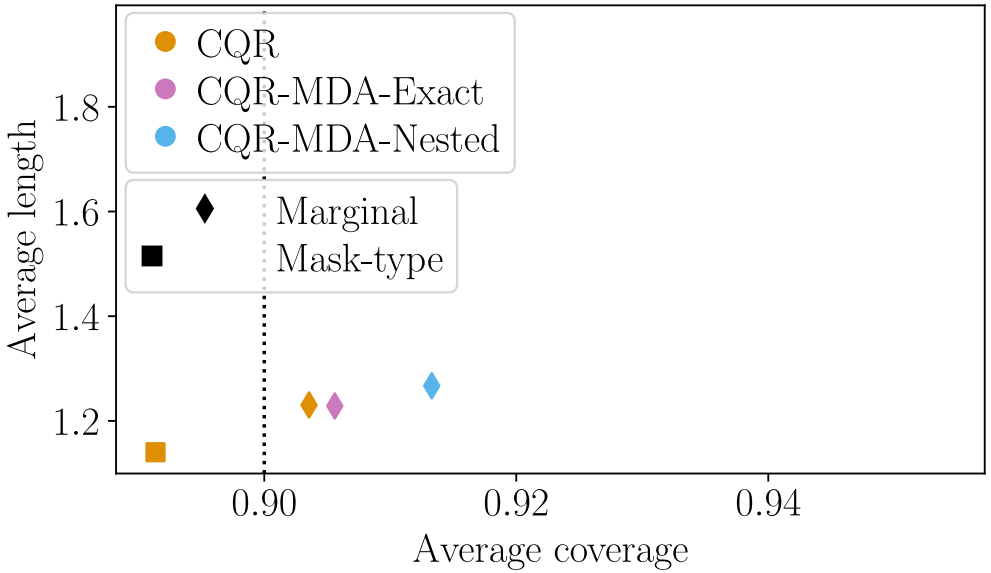
- $M_i \sim \mathcal{B}(0.2)$ for any $i \in [\![1, 3]\!]$, independently from X and ε
- $Y = X_1 \mathbb{1}\{M_1 = 0\} + 2X_1 \mathbb{1}\{M_1 = 1\} + 3X_2 \mathbb{1}\{M_2 = 1, M_3 = 1\} + \varepsilon$



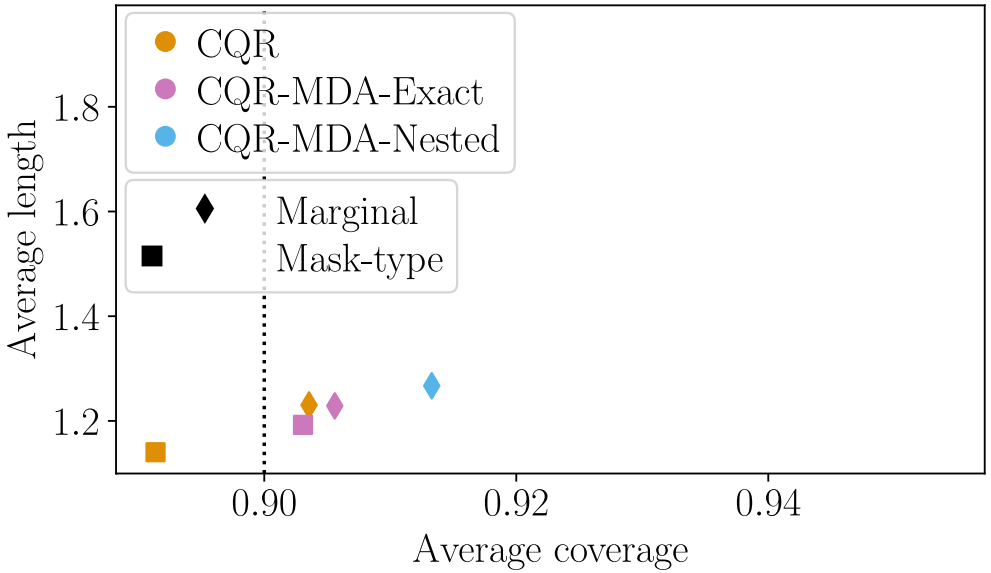
Real data experiment: TraumaBase®, critical care medicine



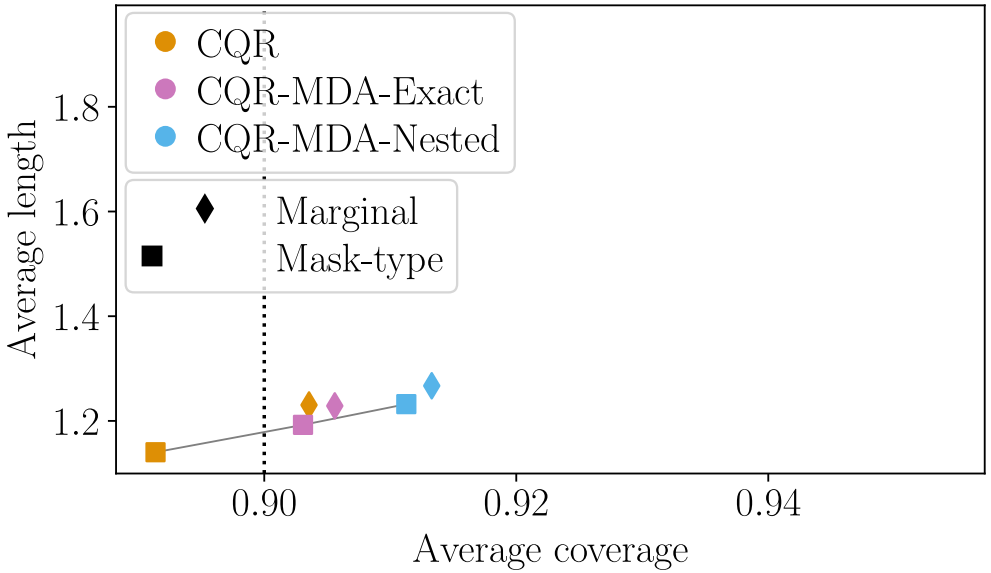
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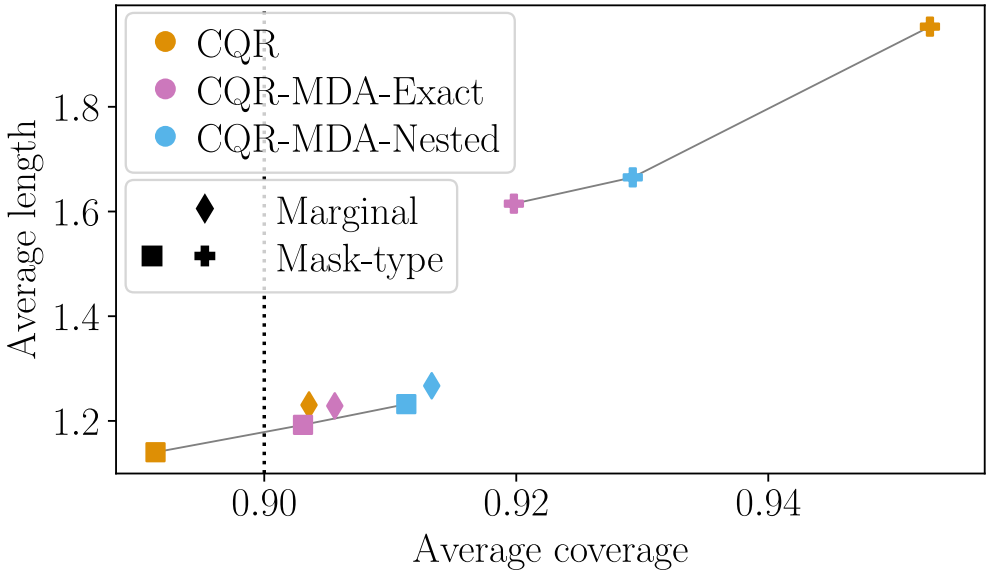
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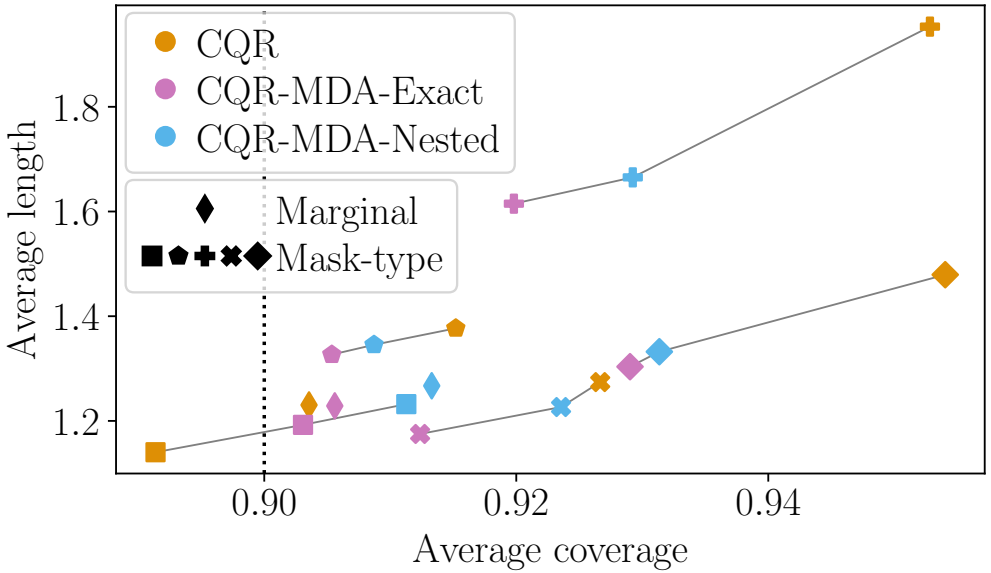
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References i

Ayme, A., Boyer, C., Dieuleveut, A., and Scornet, E. (2022). Near-optimal rate of consistency for linear models with missing values. *ICML*.

Barber, R. F., Candès, E. J., Ramdas, A., and Tibshirani, R. J. (2021). Predictive inference with the jackknife+. *The Annals of Statistics*, 49(1).

Le Morvan, M., Josse, J., Scornet, E., and Varoquaux, G. (2021). What's a good imputation to predict with missing values? *NeurIPS*.

Zhu, Z., Wang, T., and Samworth, R. J. (2019). High-dimensional principal component analysis with heterogeneous missingness. arXiv.

https://conformalpredictionintro.github.io